

Supply-chain risk language and stock market reactions: Evidence from US hardware firms

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September 2026

Abstract

Do investors respond to what managers say about supply-chain risk on earnings calls? I build text-based measures of supply-chain risk and resolution language, following Theile et al. (2026), and apply them to 11,950 earnings calls from 378 US hardware firms in fiscal years 2010 through 2019. Calls in the highest-risk quintile earn two-day abnormal returns that are 1.69 percentage points below calls in the lowest-risk quintile. With firm and calendar-quarter fixed effects, a one-standard-deviation increase in supply-chain risk language is associated with a 0.82 percentage point lower abnormal return. On the subsample with analyst forecasts, controlling for earnings surprises shrinks the magnitude of the estimate from -0.85 to -0.71 (16% smaller). Language about resolving supply problems is associated with modestly higher returns, but this relationship is weaker and less robust. Over the following 59 trading days, part of the announcement spread reverses, mainly because the lowest-risk portfolio drifts down.

1 Introduction

Hardware companies depend on long chains of specialized suppliers, and a problem at any link can stop production. In 2011, the Tōhoku earthquake disrupted automotive and electronics component supply worldwide, and flooding in Thailand cut global hard-drive output for months. Investors see these problems only partly. Financial statements record their effects after the fact, while earnings calls are one of the few recurring settings where managers describe supply conditions in their own words while they are still unfolding.

This paper asks whether that discussion moves stock prices. The answer is not obvious. If managers' supply-chain comments mostly restate what the reported numbers already show, they should carry no additional information once the numbers are known. If the comments reveal something forward-looking, such as a constraint that will limit next quarter's shipments or a fix that is already working, markets should respond at the time of the call.

To measure supply-chain discussion, I count risk words that appear near supply-chain terms in each transcript. The approach follows the proximity-based text measures of Hassan et al. (2019) and the supply-chain specific dictionaries of Theile et al. (2026). Theile et al. find that supply-chain risk language predicts lower announcement returns across a broad cross-section of US firms.

My experience in hardware product development motivated my interest in how sourcing constraints affect firms. A component may be inexpensive yet essential to production, and replacing it may require redesign, testing, or qualification. Dependencies can also extend beyond a company's direct suppliers, making disruptions difficult to anticipate. Hardware companies therefore provide a useful setting for examining whether disclosures of supply-chain risk contain information that investors associate with changes in firm value. If supply-chain language carries real information, it should matter most for these firms, so the hardware sample offers a sharper test of the mechanism.

I find that supply-chain risk language is strongly associated with lower two-day abnormal returns. The highest-risk quintile underperforms the lowest-risk quintile by 1.69 percentage points. With firm and calendar-quarter fixed effects, a one-standard-deviation increase in risk language is associated with a -0.82 percentage point return. The estimate is larger in magnitude than the -0.56 percentage points reported by Theile et al., although differences in measurement, sample, and specification prevent a direct comparison. After controlling for provider-reported earnings surprises, the estimated association remains negative and statistically significant at -0.71 percentage points, a 16.4% reduction in magnitude relative to the regression without this control on the same sample. This suggests that the negative association is not fully accounted for by the measured earnings surprise.

Language about resolving supply problems is associated with higher returns, but the evidence is weaker: it is statistically significant in baseline regressions but not in the main portfolio or after controlling for earnings surprises.

This paper relates to three literatures. Event studies such as Hendricks and Singhal (2003) show that publicly announced supply-chain disruptions reduce shareholder value. Text-based work shows that the wording of corporate disclosures carries information that prices respond to (Loughran and McDonald 2011; Hassan et al. 2019). Research on production networks shows that shocks to specialized suppliers spread to their customers (Barrot and Sauvagnat 2016). This paper connects these by asking whether routine managerial discussion, rather than discrete disruption announcements, conveys supply-chain information to investors.

2 Data and sample construction

The company screen contains 400 US-based public firms. Available transcripts cover 379 firms, and the final analysis includes 378.

The starting dataset contains 12,832 earnings-call records from Alpha Vantage. Transcripts are matched to historical issuer identities, issuer fiscal periods, and earnings-release dates. Observations require valid transcript-period information, a usable release-date mapping, complete return windows, and a historical Standard Industrial Classification (SIC) assignment.

Table 1. Construction of the analysis sample

Sample stage	Calls
Input call records	12,832
Valid issuer and transcript-period information	12,713
Usable earnings-release mappings	11,982
Complete CAR windows	11,954
Historical SIC assignments	11,950
Final sample size	11,950

The final sample contains 11,950 calls from 378 firms. Historical SIC assignments use the latest filing available by the observed fiscal-period end.

Coverage increases over the study period. Fiscal 2010 contributes 355 calls from 123 firms, compared with 1,433 calls from 368 firms in fiscal 2019. The study boundary follows issuer fiscal periods; associated earnings releases span calendar quarters from 2009Q3 through 2020Q1. The sample includes 240 fiscal-2019 observations released in 2020.

3 Measuring supply-chain risk and resolution

SCRisk measures how often supply-chain and risk terms occur near each other in an earnings-call transcript. A supply-chain term alone does not signal risk, since a manager may just be describing operations. A supply-chain term near a word signaling uncertainty or danger usually does. SCRisk therefore counts supply-chain terms that appear within ten words of a risk term. Resolution counts the subset of those pairs that also appear near a term describing mitigation or recovery.

The supply-chain vocabulary contains 254 terms. To build it, I trained word representations on 50,000 annual filings from 1997 to 2021 and selected terms whose usage closely resembled a set of seed words such as "supplier" and "sourcing." Each supply-chain term is weighted by its similarity to the seed words, so central terms count more than peripheral ones. The risk dictionary (161 terms) and the resolution dictionary (55 terms) are built primarily from the published word lists in Theile et al. (2026). Both measures are divided by transcript length, so that longer calls do not automatically score higher, and then scaled by their standard deviation. Coefficients therefore read as the association with a one-standard-deviation change.

The measures are sparse. About 28% of calls have zero SCRisk and 83% have zero Resolution, because many calls discuss supply chains without pairing them with risk language. Appendix A describes the vocabulary construction, dictionary reconciliation, and scoring rules in full.

4 Abnormal returns and portfolio construction

Event returns

The event date is the earnings release. Day 0 is the first factor-calendar trading day on or after the release, without an after-hours adjustment.

Expected excess returns are estimated using the Carhart (1997) four-factor model, comprising market, size, value, and momentum factors with an intercept. Each event requires 200 complete daily observations at trading offsets -209 through -10.

Daily returns use adjusted closing prices. Abnormal returns are realized returns minus fitted expected returns. CAR(0,1) sums abnormal returns over two trading days, while CAR(2,60) sums returns over the following 59 trading days. The common analysis sample requires both windows to be complete.

Fractional portfolios

SCRisk, Resolution, and both CAR measures are winsorized at the sample's 1st and 99th percentiles. SCRisk portfolios are formed within historical SIC divisions over the pooled study period and then combined across divisions using call weights.

The sample contains 3,326 zero-SCRisk calls and 9,962 zero-Resolution calls, representing 27.83% and 83.36% of observations, respectively. Fractional allocation distributes a tied-score group proportionally across any quintile boundaries it spans. Every call within that group receives the same allocation, and each call's membership weights sum to one.

Each pooled risk portfolio has fractional mass 2,390, defined as the sum of its membership weights. Resolution is subsequently sorted within each division and risk portfolio using those weights. Each nested cell has fractional mass 478.

Confidence intervals are clustered by firm and account for covariance between portfolios that share observations.

5 Portfolio results

Table 2. Mean cumulative abnormal returns by SCRisk portfolio

SCRisk portfolio	CAR(0,1) (%)	CAR(2,60) (%)
Q1	1.131	-0.929
Q2	0.432	-0.339
Q3	0.856	-0.398
Q4	0.314	0.020
Q5	-0.557	-0.227

Mean $CAR(0,1)$ is 1.13% in the lowest-risk portfolio and -0.56% in the highest. The highest-minus-lowest difference is -1.687 percentage points, with a 95% confidence interval of [-2.075, -1.300].

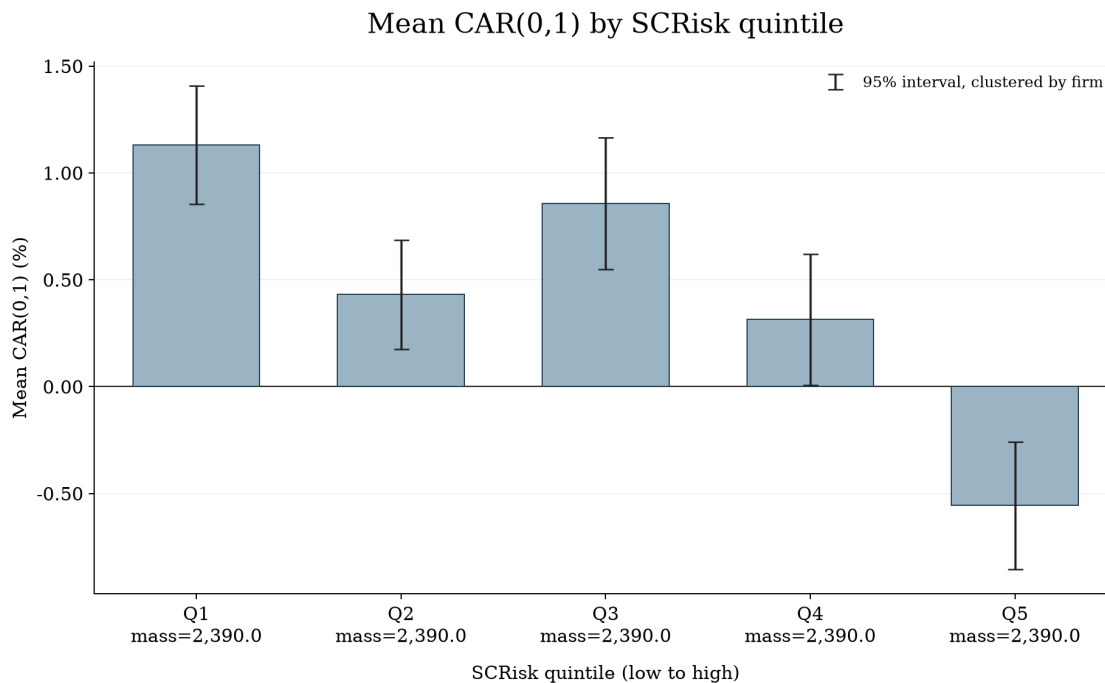


Figure 1. Mean $CAR(0,1)$ by SCRisk quintile. Portfolios are formed within SIC divisions using fractional allocation. The sample contains 11,950 calls from 378 firms. Bars show firm-clustered 95% confidence intervals.

The relationship is strong at the extremes but not monotonic in between: the middle quintile (0.86%) earns more than the second (0.43%). The middle quintiles are not statistically distinguishable from one another; their confidence intervals overlap. The robust pattern is the gap between calls with no supply-chain risk language (Q1, all zero scores) and the highest-risk calls.

The highest-minus-lowest Resolution portfolio difference is +0.138 percentage points, with a confidence interval of [-0.134, 0.409]. Figures 2 and 3 show the pooled Resolution portfolios and the joint distribution of returns across SCRisk and Resolution portfolios.

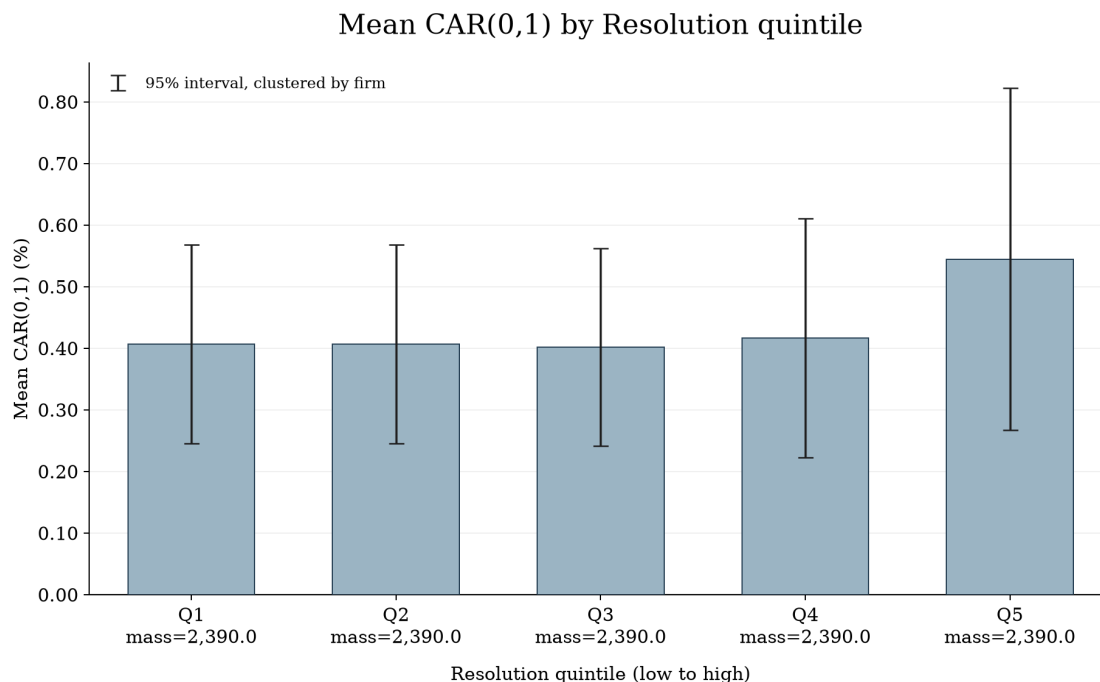


Figure 2. Mean CAR(0,1) by Resolution quintile. Resolution is sorted within SIC divisions and SCRisk portfolios, after which corresponding portfolios are pooled. Bars show firm-clustered 95% confidence intervals.

Resolution sorts produce a positive but statistically insignificant spread. Because 83% of calls have zero Resolution, the lower quintiles are made up entirely of tied zero scores, which limits what a quintile sort can detect. The regressions in Section 6, which use the continuous measure, are better suited to this variable.

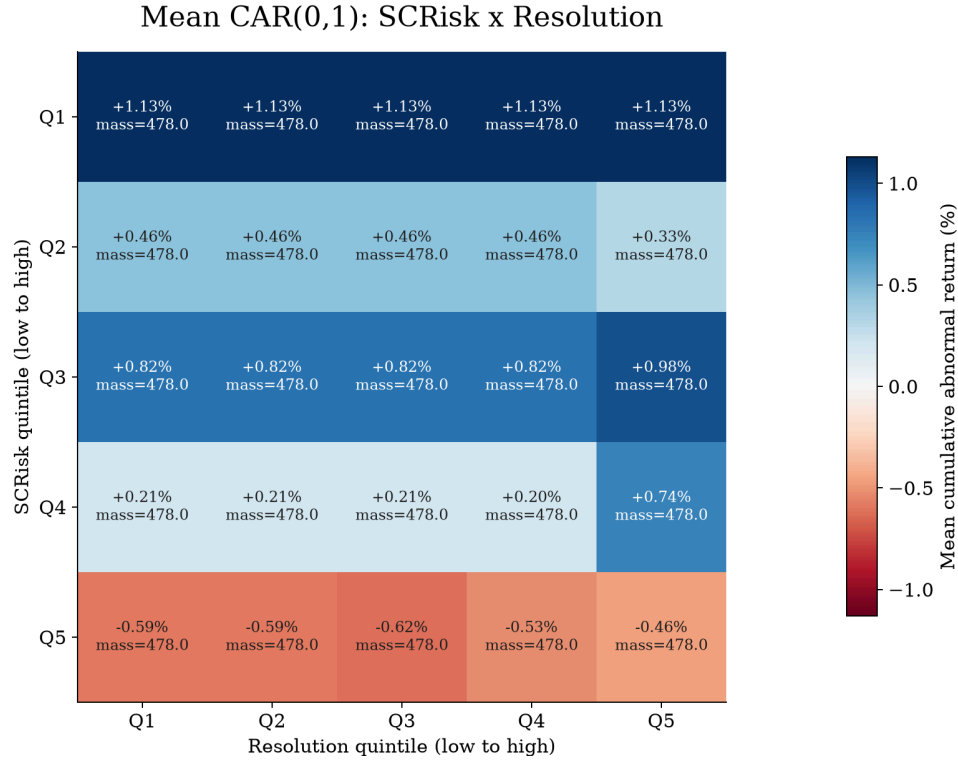


Figure 3. CAR(0,1) across SCRisk and Resolution portfolios. Each cell reports the mean two-day abnormal return from the nested sorts and has fractional mass 478.

Because Resolution is counted only within supply-chain risk pairs, calls with zero SCRisk also have zero Resolution. The Q1 row is therefore identical across columns, and in most rows only the highest Resolution column differs from the others. The informative comparison is within each row: in the third and fourth SCRisk quintiles, calls with the most resolution language earn higher returns (0.98% vs. 0.82% and 0.74% vs. 0.21%), and the pattern is weaker in the highest-risk row (−0.46% vs. −0.59%) and reversed in the second (0.33% vs. 0.46%). Cell-level confidence intervals are not reported, so these differences are descriptive.

For CAR(2,60), the highest-minus-lowest SCRisk difference is +0.702 percentage points, with a confidence interval of [0.010, 1.394]. Figures 4 and 5 present the later-window portfolio results.

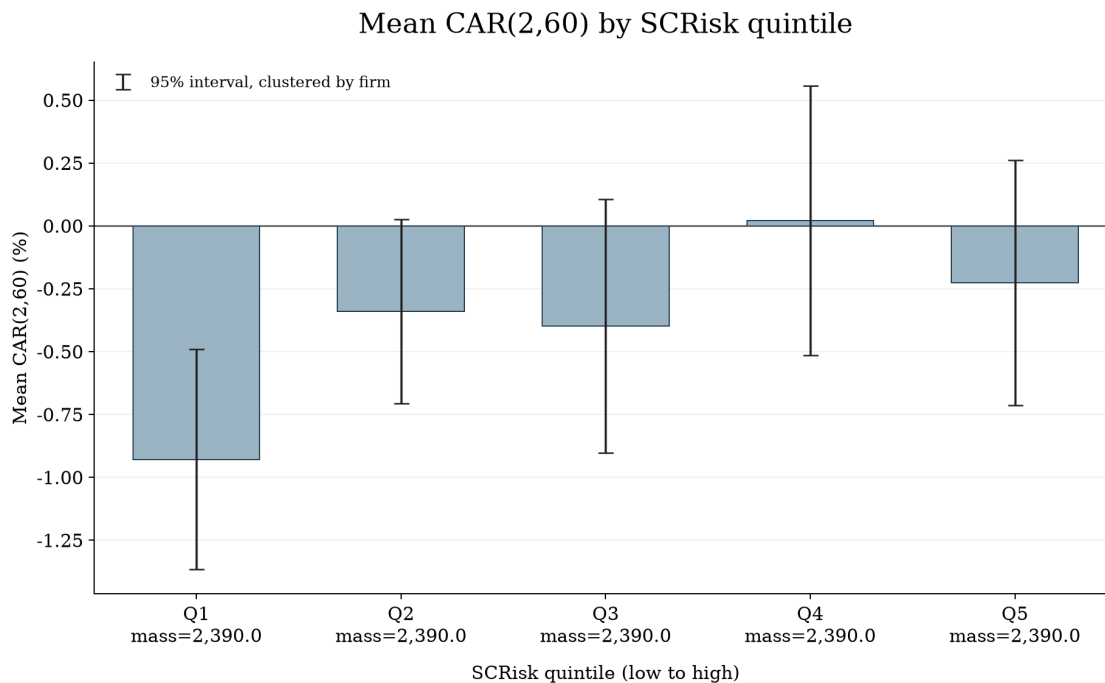


Figure 4. Mean CAR(2,60) by SCRisk quintile. Returns cover trading days 2–60 using the same sample and portfolio assignments as the short-window analysis.

Over the following 59 trading days, the pattern partly reverses: the highest-minus-lowest difference is +0.70 percentage points. The reversal comes mainly from the lowest-risk portfolio, whose later-window return of -0.93% is about 0.7 percentage points below the average of the other four quintiles. Abnormal returns are negative in four of the five quintiles, averaging -0.37% , which suggests some drift in this sample that the four-factor model does not capture. The pattern is therefore best read as Q1 underperforming the other portfolios rather than high-risk firms recovering. One possibility is that it reflects the announcement return itself rather than supply-chain language: Q1 has the largest two-day return, and large announcement reactions may partly reverse regardless of what managers say.

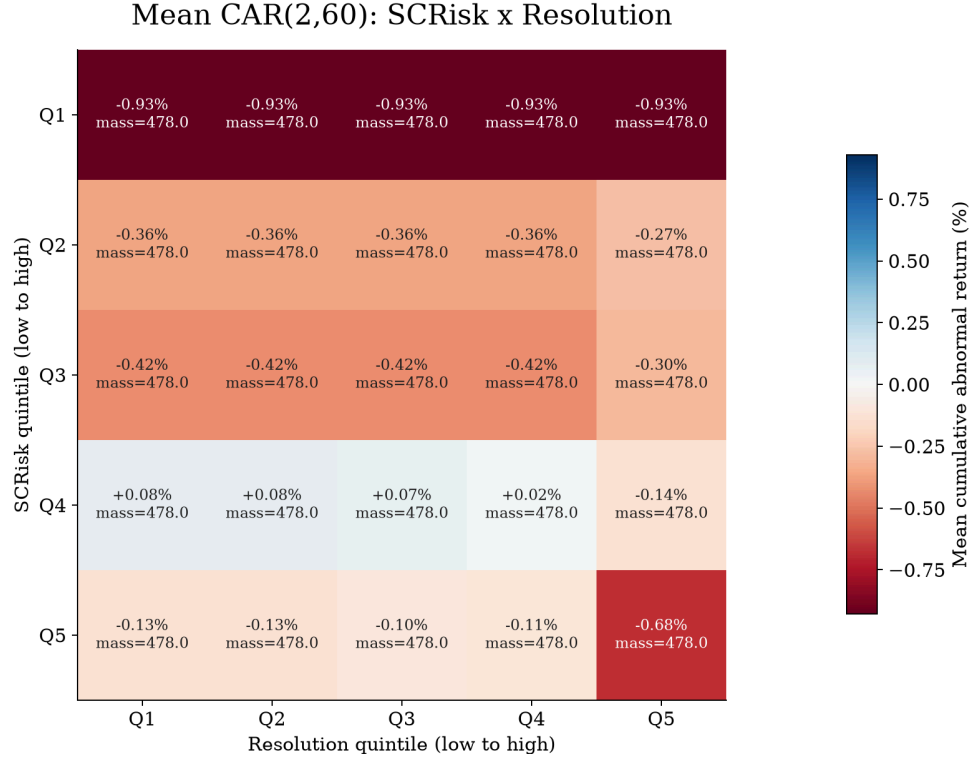


Figure 5. CAR(2,60) across SCRisk and Resolution portfolios. Cells report mean abnormal returns over trading days 2–60 from the nested portfolio sorts.

As with Figure 3, variation across Resolution columns is limited by construction, and the later-window cell differences are not individually tested.

6 Regression results

The regression analysis examines the relationship between risk language and CAR after accounting for firm characteristics and calendar conditions. The primary specification is:

$$100 * CAR_{i,c}(0,1) = \alpha_i + \delta_{q(c)} + \beta SCRisk_{i,c} + \gamma Resolution_{i,c} + \varepsilon_{i,c} \quad (1)$$

Here, i identifies the firm, c the call, and $q(c)$ the calendar year-quarter of the event trading date. Firm fixed effects (α_i) capture persistent differences between companies, while calendar-quarter fixed effects ($\delta_{q(c)}$) capture conditions shared across firms in the same period. β and γ represent the estimated relationships of SCRisk and Resolution with returns. $\varepsilon_{i,c}$ represents the model's error term, which is the part of a call's return that's not accounted for by any of the aforementioned variables.

I estimate three main specifications: SCRisk alone (M1), SCRisk with Resolution (M2), and both measures with firm and calendar-quarter fixed effects (M3). M1–M3 use the same 11,950 calls with equal observation weights. Table 3 also reports M3 excluding fiscal 2010 and, on the subsample with earnings-surprise data, M3 without (M4) and with (M5) a surprise control. Standard errors are clustered by firm. Coefficients are expressed in CAR percentage points per standard deviation.

Table 3. Regressions of CAR(0,1) on SCRisk and Resolution

Variable	M1	M2	M3	Excluding FY2010	M4	M5
Description	SCRisk Alone	SCRisk w/ Resolution	SCRisk w/ Resolution, plus firm and calendar-quarter fixed effects	M3 excluding FY2010	M3 on the earnings surprise sample	M4 plus earnings surprise
SCRisk	-0.6183	-0.7123	-0.8227	-0.8386	-0.8511	-0.7118
Clustered SE	(0.0830)	(0.0882)	(0.1065)	(0.1098)	(0.1498)	(0.1405)
95% CI	[-0.782, -0.455]	[-0.886, -0.539]	[-1.032, -0.613]	[-1.054, -0.623]	[-1.147, -0.555]	[-0.989, -0.435]
Resolution	—	0.2326	0.2233	0.2266	+0.3111	+0.1976
Clustered SE	—	(0.1020)	(0.1035)	(0.1066)	(0.1534)	(0.1342)
95% CI	—	[0.032, 0.433]	[0.020, 0.427]	[0.017, 0.436]	[0.008, 0.614]	[-0.067, 0.463]
Firm fixed effects	No	No	Yes	Yes	Yes	Yes
Calendar-quarter fixed effects	No	No	Yes	Yes	Yes	Yes
Earnings surprise control	No	No	No	No	No	Yes
# of calls	11,950	11,950	11,950	11,595	5,476	5,476
Firms	378	378	378	378	171	171
Overall R ²	0.0054	0.0059	0.0562	0.0563	0.0650	0.1571
Partial R ² after both fixed effects	—	—	0.0064	0.0066	0.0072	0.1051

Notes: The outcome is 100 times winsorized CAR(0,1). Score scaling and winsorization thresholds are held constant across specifications. Confidence intervals use firm-clustered standard errors with a finite-sample correction and a t distribution with 377 degrees of freedom for M1-M3. M4 and M5 have 170 degrees of freedom. Overall R² includes the fixed effects. Partial R² measures the additional explanatory contribution of the non-fixed-effect regressors after removing both fixed effects; in M5 this includes the earnings surprise.

The SCRisk coefficient grows in magnitude as controls are added, from -0.62 in the pooled regression to -0.82 with fixed effects. Adding firm fixed effects means the comparison is within a company: the same firm earns lower returns in quarters when it discusses supply-chain risk more than usual. The larger coefficient suggests that firms that habitually discuss supply chains, perhaps because their operations are more complex, do not earn systematically lower returns.

Including Resolution increases the SCRisk coefficient because the two measures are positively correlated. Managers who describe a supply problem often also describe their response. Without Resolution in the model, the positive effect of resolution language partly offsets the negative effect of risk language.

Excluding fiscal 2010, the year with the thinnest coverage, leaves both coefficients nearly unchanged.

A natural concern is that supply-chain risk language may accompany disappointing earnings. To address this, I add the earnings surprise reported by Alpha Vantage, defined as actual minus estimated EPS, scaled by the last unadjusted closing stock price before earnings release and winsorized at 1st and 99th percentiles. Surprise data are available for 5,476 of the 11,950 calls, so M4 re-estimates the primary specification on this subsample. The SCRisk coefficient moves from -0.85 (SE 0.150) to -0.71 (SE 0.140), a 16.4% reduction in magnitude, and remains statistically significant. Most of the estimated association between supply-chain risk language and returns therefore persists after controlling for this measure of earnings surprise. The Resolution coefficient falls from 0.31 to 0.20 and is no longer statistically significant ($p = 0.143$), consistent with some overlap between resolution language and favorable earnings news.

7 Limitations

The estimates are associations, not causal effects. Supply-chain language is chosen by managers, who may discuss risks more candidly when results are already poor or less candidly when they want to reassure investors. The earnings-surprise control addresses the most direct version of this concern, but guidance revisions and other news released at the same time may still affect both language and returns. The control is available only for the 5,476 calls with analyst EPS estimates, a subsample that likely overrepresents larger, more widely followed firms, so the controlled estimate may not generalize to the full sample.

The sample is selected on current listing characteristics, so firms that were delisted, acquired, or went bankrupt during 2010–2019 are underrepresented. If severe supply disruptions contributed to some of those failures, the most negative high-risk calls are missing, and the estimates likely understate the true relationship. Acquisitions could push in the other direction, so the net bias is uncertain.

Earnings-release dates are matched to trading days without an after-hours adjustment. For calls held after the market closes, part of the reaction falls on day 1, which the two-day window captures. Standard errors are clustered by firm but do not account for shocks shared across firms in the same quarter beyond the fixed effects. Finally, the vocabulary was trained on filings through 2021, after the sample period ends, so the measure uses language conventions that investors at the time could not have known.

8 Conclusion

Supply-chain risk language on earnings calls is associated with meaningfully lower announcement returns among US hardware firms. The relationship holds within firms and across calendar quarters and survives controls for earnings surprises. Resolution is significant in the baseline regressions, but not after adding earnings surprises. The later partial reversal, concentrated among low-risk calls, but this pattern is imprecise and may reflect the size of the announcement reaction rather than the content of the call.

A natural next question is which firms' disclosures matter most. If supply-chain language matters because disruptions are costly, the effect should be largest for firms that cannot easily replace a critical supplier. Testing this requires a measure of supplier dependence based on firms' own disclosures, which is the direction of my ongoing work.

Contribution and assistance

I selected the company-sample focus and directed choices about portfolio scope, dictionary specifications, and the treatment of tied scores. AI assistants supported coding, source collection, and validation. The empirical framework builds on Theile et al. (2026).

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Appendix A: Measure construction and portfolio details

A.1 Supply-chain vocabulary

The supply-chain vocabulary was built from a year-balanced corpus of 50,000 annual filings published from 1997 to 2021. I represented each word by the contexts in which it appears, using positive pointwise mutual information. I then reduced these representations with truncated singular value decomposition, so that words used in similar contexts have similar vectors. Candidate terms were ranked by cosine similarity to a set of supply-chain seed terms such as "supplier," "sourcing," and "procurement".

Final selection used three criteria: consistency with the published anchor terms in Theile et al. (2026), semantic relevance to supply chains, and stability across model specifications and corpus subsamples. Vocabulary overlap was checked using a different random seed and a 50% corpus subsample; the selected model's Jaccard similarities were 0.864 and 0.791. The final vocabulary contains 254 terms. Each term's weight is its maximum cosine similarity to the seed vocabulary.

A.2 Risk and resolution dictionaries

The risk dictionary contains 161 terms: 144 word forms published by Theile et al. (2026) and 17 additional candidates. The 17 candidates were terms in Hassan et al.'s published appendix that were absent from Theile's displayed list, consistent with Theile's reported count of unobserved dictionary terms. The resolution dictionary contains 55 terms, combining the published anchor terms with inflected forms of those anchors, such as plurals and verb tenses.

A.3 Scoring

Transcripts are scored by exact-term matching. For each supply-chain term occurrence, I identify every qualifying supply-chain/risk occurrence pair. If a risk term falls within ten tokens of a supply-chain term, the pair qualifies and contributes the supply-chain term's weight. SCRisk is the sum of qualifying weights divided by the transcript's token count. Resolution uses the same qualifying pairs and counts a pair only when a resolution term also appears within ten tokens of the supply-chain occurrence. It is likewise normalized by token count.

Each raw measure is divided by its population standard deviation across the 12,713 transcripts with valid issuer and period information, without mean centering. Scaling is done before return-based sample exclusions, so the scale does not depend on return data availability. A score of zero means the transcript contains no qualifying pairs under these rules.

A.4 Fractional portfolio allocation

Because 27.83% of calls have zero SCRisk and 83.36% have zero Resolution, many calls share identical scores, and conventional quintile breakpoints would assign tied calls to portfolios arbitrarily. I instead allocate each group of tied calls proportionally across every quintile boundary the group spans. Each call in the group receives the same fractional membership weights, and each call's weights sum to one. As a result, each SCRisk portfolio has a total fractional mass of 2,390 calls.

SCRisk portfolios are formed within historical SIC divisions over the pooled sample period and then combined across divisions using call weights. Resolution portfolios are then formed within each division-by-SCRisk portfolio, using the same fractional method, so each of the 25 nested cells has a fractional mass of 478. Confidence intervals are clustered by firm and account for covariance between portfolios that share fractionally allocated calls.